

Q: Suppose we center / scale the covariates, will the MLE/OLS estimators of β_j change?

A. Centering

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i \\ &= (\beta_0 + \beta_1 \bar{x}_1 + \cdots + \beta_p \bar{x}_p) + \beta_1 (x_{i1} - \bar{x}_1) + \cdots + \beta_p (x_{ip} - \bar{x}_p) + \varepsilon_i \\ &= \alpha_0 + \beta_1 x_{i1}^c + \cdots + \beta_p x_{ip}^c + \varepsilon_i \end{aligned}$$

$$X_c = \begin{bmatrix} \mathbf{1} & x_{(1)}^c & \cdots & x_{(p)}^c \end{bmatrix} \quad X = \begin{bmatrix} \mathbf{1} & x_{(1)} & \cdots & x_{(p)} \end{bmatrix}$$

where

$$x_{(j)}^c = x_{(j)} - \bar{x}_j \mathbf{1}, \quad x_{(j)} = \begin{bmatrix} x_{1j} \\ \vdots \\ x_{nj} \end{bmatrix}.$$

$$\hat{\beta}_{\text{OLS}}^c = (X_c^\top X_c)^{-1} X_c^\top Y, \quad \hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top Y.$$

$$X_c = X \begin{bmatrix} 1 & -\bar{x}_1 & \cdots & -\bar{x}_p \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = X \begin{bmatrix} 1 & -(\bar{x}_1 \cdots \bar{x}_p) \\ \mathbf{0} & I_p \end{bmatrix} = XA.$$

$$\begin{aligned} \Rightarrow X_c \underline{a} &= XA \underline{a} = X \underline{b} \Rightarrow e(X_c) = C(X), \\ X \underline{a} &= X_c A^{-1} \underline{a} = X_c \underline{d}. \end{aligned}$$

$\Rightarrow$ Orthogonal projection matrix is the same:

$$P(X_c) = P(X).$$

$$[\text{if } X \text{ is of full column rank}] \Rightarrow X_c (X_c^\top X_c)^{-1} X_c^\top = X (X^\top X)^{-1} X^\top$$

$$\Rightarrow ZXA(X_c^\top X_c)^{-1} X_c^\top Y = ZX(X^\top X)^{-1} X^\top Y.$$

Suppose Z is such that

$$ZX = I_{p+1} \quad (\text{i.e., the left-inverse of } X) \Rightarrow A\hat{\beta}_{\text{OLS}}^c = \hat{\beta}_{\text{OLS}}.$$

The k -th component:

For $k = 2, \dots, p+1$:

$$\underline{e}_k^\top A \hat{\beta}_{\text{OLS}}^c = \hat{\beta}_{k,\text{OLS}} \Rightarrow \hat{\beta}_{k,\text{OLS}}^c = \hat{\beta}_{k,\text{OLS}},$$

where

$$\underline{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k\text{th.}$$

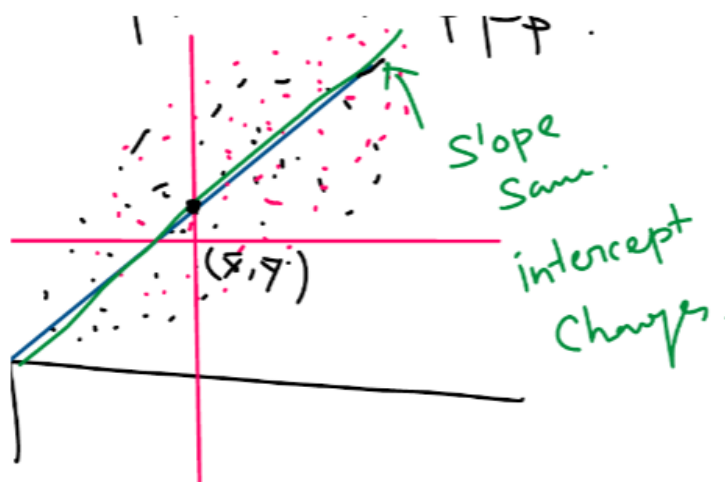
$$\underline{e}_k^\top A = (0 \dots 0 \ 1 \ 0 \dots 0) A = \underline{e}_k^\top.$$

Also,

$$\begin{aligned} \underline{e}_1^\top A \hat{\beta}_{\text{OLS}}^c &= \hat{\beta}_0 \\ \Rightarrow (1 \ -\bar{x}_1 \ \dots \ -\bar{x}_p) \begin{pmatrix} \hat{\beta}_0^c \\ \hat{\beta}_1^c \\ \vdots \\ \hat{\beta}_p^c \end{pmatrix} &= \hat{\beta}_0. \end{aligned}$$

$$\Rightarrow \hat{\beta}_0 = \hat{\beta}_0^c + \bar{x}_1 \hat{\beta}_1^c + \dots + \bar{x}_p \hat{\beta}_p^c = \hat{\beta}_0 + \bar{x}_1 \hat{\beta}_1 + \dots + \bar{x}_p \hat{\beta}_p.$$

$$X_c = [\mathbf{1} \ x_{(1)}^c \ \dots \ x_{(p)}^c] = [\mathbf{1} \ Z_c].$$



$$\mathbf{1}^\top x_{(j)}^c = 0 \Rightarrow \mathbf{1}^\top (x_{(j)} - \bar{x}_j \mathbf{1}) = \sum_{i=1}^n x_{ij} - n\bar{x}_j = 0.$$

$$X_c^\top X_c = \begin{bmatrix} \mathbf{1}^\top \\ Z_c^\top \end{bmatrix} \begin{bmatrix} \mathbf{1} & Z_c \end{bmatrix} = \begin{bmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & Z_c^\top Z_c \end{bmatrix},$$

$$(X_c^\top X_c)^{-1} = \begin{pmatrix} \frac{1}{n} & \mathbf{0}^\top \\ \mathbf{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix}.$$

$$P(X_c) = X_c(X_c^\top X_c)^{-1}X_c^\top = \begin{bmatrix} \mathbf{1} & Z_c \end{bmatrix} \begin{pmatrix} \frac{1}{n} & \mathbf{0}^\top \\ \mathbf{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix} \begin{bmatrix} \mathbf{1}^\top \\ Z_c^\top \end{bmatrix}$$

(verify)

$$= \frac{1}{n} \mathbf{1} \mathbf{1}^\top + P(Z_c) = P(\mathbf{1}) + P(Z_c).$$

Sum of two orthogonal projection matrices, which are orthogonal to each other.

$$H_n = I - \frac{1}{n} \mathbf{1} \mathbf{1}', \quad \mathbf{1}(\mathbf{1}^\top \mathbf{1})^{-1} \mathbf{1}'.$$

Residual:

$$Y^\top (I - P(X_c)) Y = Y^\top \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}' - P(Z_c) \right) Y$$

$$= Y^\top (I - P(Z_c)) Y - n \bar{Y}^2.$$

Scaling(Boston Housing Data)

$$X_s = \begin{bmatrix} \mathbf{1} & x_{(1)}/s_1 & \dots & x_{(p)}/s_p \end{bmatrix} = X \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1/s_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1/s_p \end{bmatrix} = X D,$$

where

$$s_j = \text{SD}(x_{1j}, \dots, x_{nj}), \quad D = \text{Diag}\left(1, \frac{1}{s_1}, \dots, \frac{1}{s_p}\right).$$

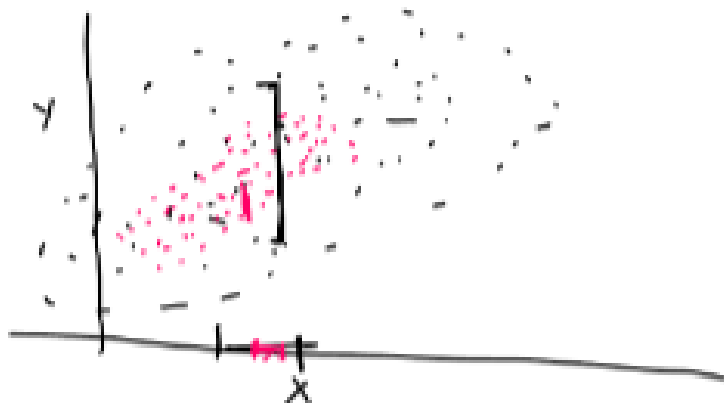
$$P(X_s) = X_s(X_s^\top X_s)^{-1}X_s^\top = X D (D X^\top X D)^{-1} D X^\top = P(X),$$

$$e(X_s) = C(X) \iff P(X_s) = P(X) \quad [\text{Same } \hat{Y}, e, \text{RSS}].$$

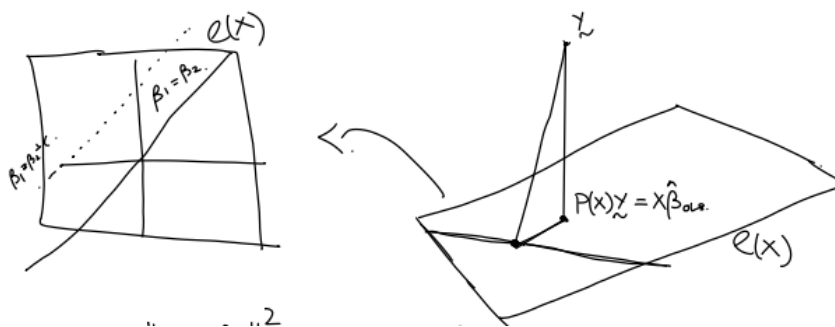
$$D^{-1} \hat{\beta}_{\text{OLS}} = \hat{\beta}_{\text{OLS}}^s \quad [\text{assume } (X^\top X) \text{ exists}].$$

$$\hat{\beta}_{\text{OLS}} = D \hat{\beta}_{\text{OLS}}^s, \quad \hat{\beta}_j^s = s_j \hat{\beta}_j, \quad D^{-1} = \text{Diag}(1, s_1, \dots, s_p).$$

Iris Data.



Finding MLE/OLS estimate of β under some linear constraint



As

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{OLS}\|^2 + \|X\hat{\beta}_{OLS} - X\beta\|^2.$$

Enough to find the β satisfying $A\beta = c$, such that $X\beta$ is closest to $X\hat{\beta}_{OLS}$.

MLE solution

$$\ell_n(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2.$$

Constraint:

$$A\beta = c \quad \text{where } A \text{ is an } r \times (p+1) \text{ matrix with full row rank.}$$

Lagrangian:

$$g(\beta, \sigma^2, \lambda) = \ell_n(\beta, \sigma^2) - \lambda^\top (A\beta - c), \quad \lambda \in \mathbb{R}^r.$$

$$\frac{\partial g}{\partial \beta} = 0, \quad \frac{\partial g}{\partial \sigma^2} = 0, \quad \frac{\partial g}{\partial \lambda} = 0.$$